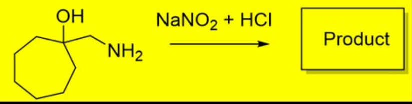
CHM102 Basic Organic Chemistry

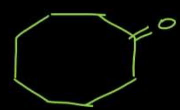
Reactive intermediates and reactivity aspects

Problem Set 4

Q2. Identify the product and and the reactive intermediate(s) involved in the reactions. Also, indicate arrow-push mechanism for the product formation.

(e)





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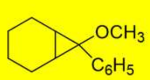
CHM102 Basic Organic Chemistry

Reactive intermediates and reactivity aspects

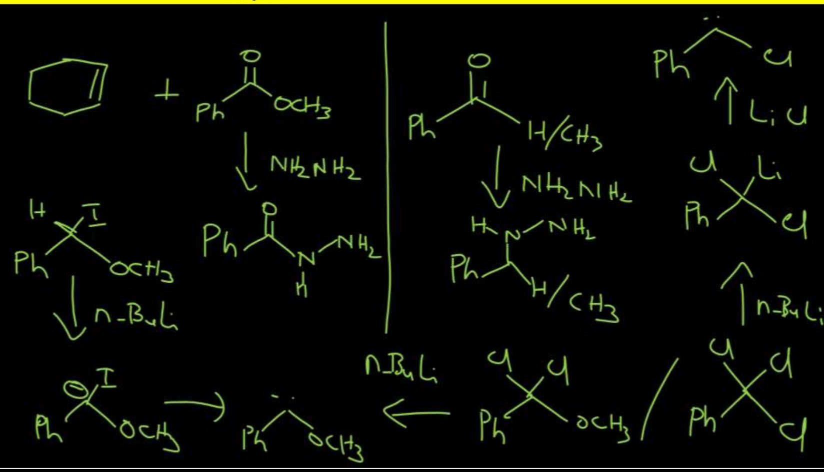
Problem Set 4

Q3. Provide a synthetic route for accessing the following compounds. Consider any starting material, reactive intermediate(s) and indicate appropriate reaction conditions.

(a)



(b)



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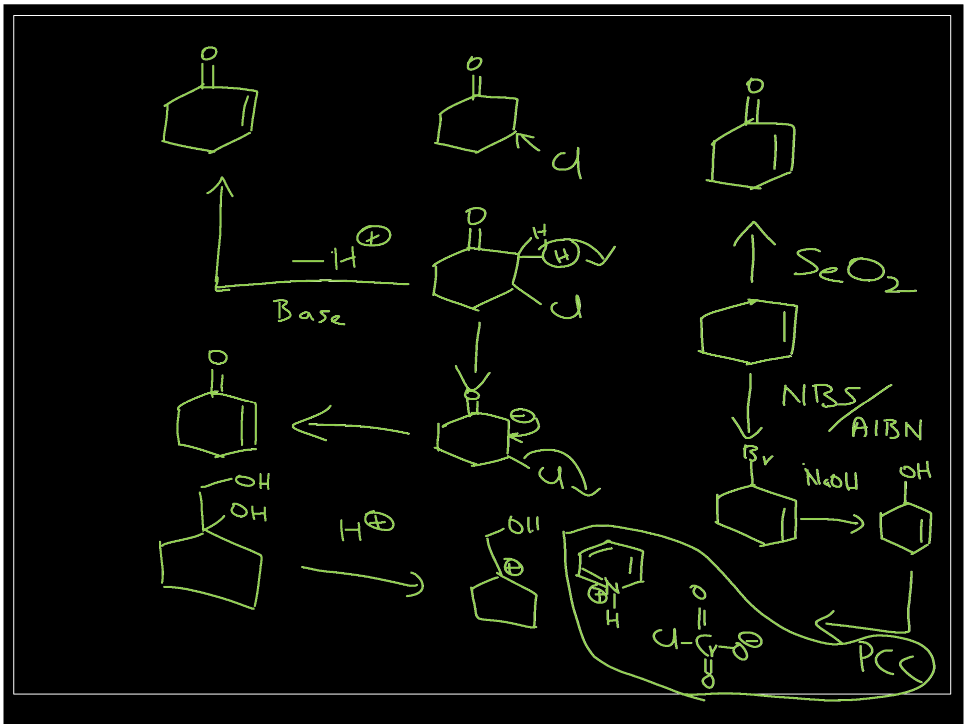

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CHM102 Basic Organic Chemistry

Reactive intermediates and reactivity aspects

Problem Set 4

Q6. Predict the major product for the following reaction, classify the reaction based on the mechanism and indicate the reason.

Reactant	Reaction condition	Product	Type of mechanism	Reason
Br	NaCN	CN	S_N2	Good nucleophile Not strong base
Br	$NaOCH_3$	OMe	S_N2	Good nucleophile + strong base
Br	KO-$^tBu_3$		E2	Nucleophilic + hindered base
Br	NaI	I	S_N2	Good nucleophile Not strongly basic
Br	$NaOCH_3$		E2	Strong base/nucleophilic
Br	EtOH	OEt +	S_N1 or E1	Polar solvent + Weak nucleophile
Br	CH_3SNa	SCH_3	S_N2	Good/weakly basic nucleophilic
Br	$NaOCH_3$		E2	Strong basic nucleophile
Br	EtOH	OEt or	S_N1 or E1	Polar solvent/weak base & nucleophile
Br	$NaOCH_3$		E2	Strong base

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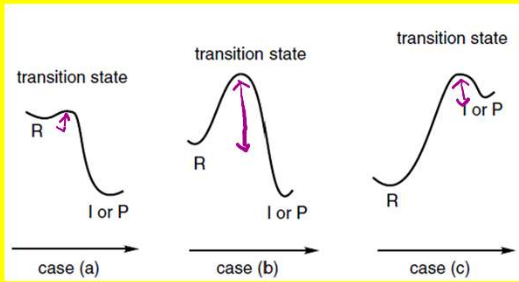
CHM102 Basic Organic Chemistry
Stereochemistry

Problem Set 3

Q13. Account for the structure of the transition state for the following cases based on the energy profiles of the reactions (R = Reactant; I = Intermediate; P = Product):



Early TS (TS close to "R")

Early TS (TS close to "R")

Late TS (TS close to "I" or "P")

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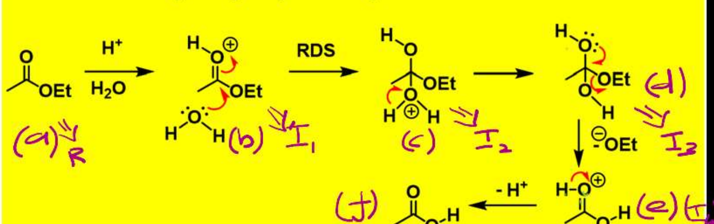
Stereochemistry

Problem Set 3

Q14. For the following reaction, draw an energy profile diagram (based on the available data) and account for the enthalpy/entropy control in the rate determining step.

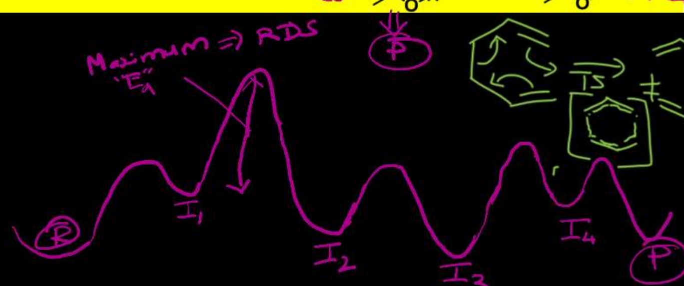
Acid-catalyzed hydrolysis of ethyl acetate in water



(b) ⇒ ⊕ive charge at "O" Higher energy than "R"

(c) ⇒ I_2 has lower in energy due to more no. of atoms compared to I_1

(d) ⇒ I_3 is more stable than I_2 due to no extra charge (all Valency satisfied)



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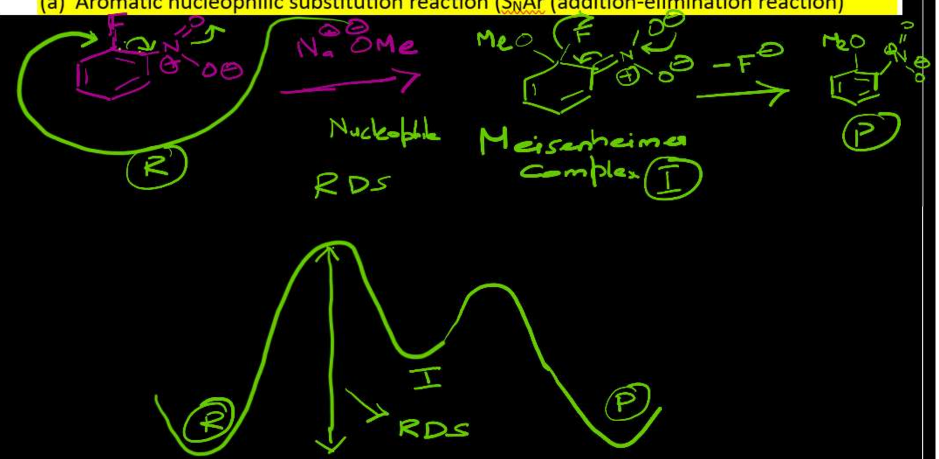
Stereochemistry

Problem Set 3

Q13. Construct energy profile diagram based on the known reaction mechanism for the following: (Also, indicate the rate determining step in each reaction)

(a) Aromatic nucleophilic substitution reaction (S_NAr (addition-elimination reaction)



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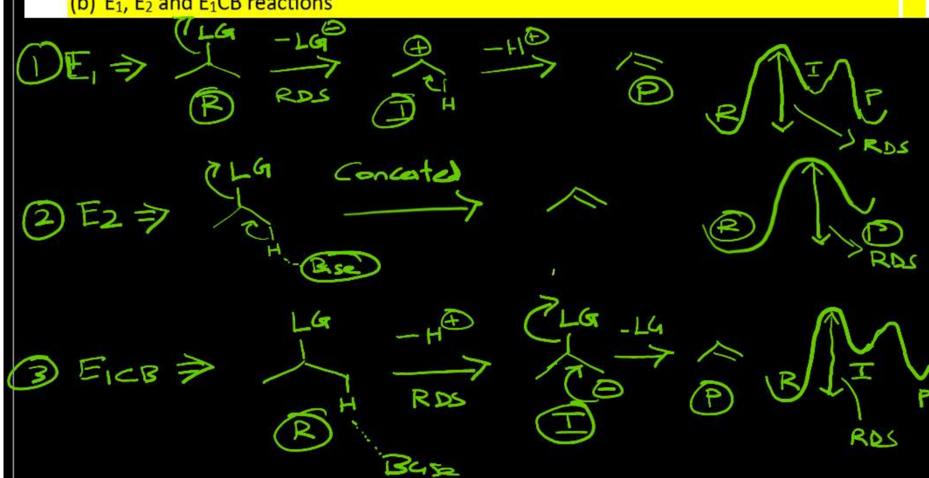
Stereochemistry

Problem Set 3

Q13. Construct energy profile diagram based on the known reaction mechanism for the following: (Also, indicate the rate determining step in each reaction)

(b) E_1, E_2 and E_1CB reactions



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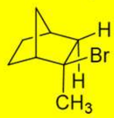
CHM102 Basic Organic Chemistry
Stereochemistry

Problem Set 3

Q14. Indicate the dominant factor among enthalpy and entropy in controlling the following reactions:

(a) Thermal decomposition of benzoyl peroxide
(b) Dimerization of cyclopentadiene (through 4+2 cycloaddition / Diels-Alder reaction)



(c) Base mediated reaction

(a) $C_6H_5-\overset{O}{\overset{\|}{C}}-O-O-\overset{O}{\overset{\|}{C}}-Ph \xrightarrow[\text{Homolysis}]{\Delta} 2\, C_6H_5\overset{O}{\overset{\|}{C}}-O\cdot$

(Single reactant) Weakest bond

$\xrightarrow{-CO_2}$ $2\, C_6H_5\cdot$

Overall ↓ rxn

$(2 \times C_6H_5\cdot) + (2 \times CO_2)$

No. of product increases ⇒ Entropy-driven rxn.

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CHM102 Basic Organic Chemistry
Stereochemistry

Problem Set 3

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(a) Thermal decomposition of benzoyl peroxide
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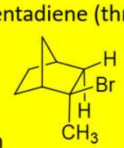
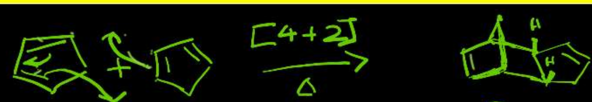


(c) Base mediated reaction

(b)



$\underbrace{4\pi e^{\ominus} \quad 2\pi e^{\ominus}}$

4+2 ⇒ Diels-Alder reaction

(2 reactant)

Product
(Single product)

Enthalpy-driven reaction

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Stereochemistry

Problem Set 3

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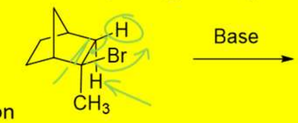
(a) Thermal decomposition of benzoyl peroxide
(b) Dimerization of cyclopentadiene (through 4+2 cycloaddition / Diels-Alder reaction)



(c) Base mediated reaction

(c)

1 - reactant

(Syn elimination)

2 - Products

(Entropy - driven rxn.)

Syn facial

(in rigid bicyclic systems syn elimination is favoured)

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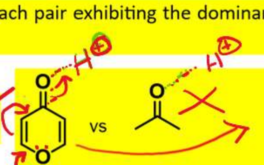
Stereochemistry

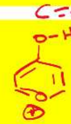
Problem Set 3

Q14. Identify the molecule among each pair exhibiting the dominant property indicated for each case and indicate the reason.

(a) Basicity



$C=O \Rightarrow$ Protonation sites

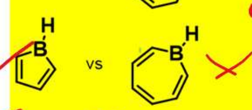


(Aromatic) stabilization

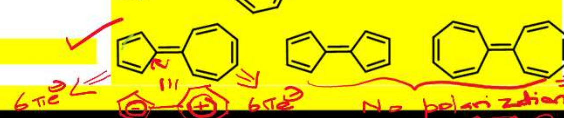
(b) Ionization of the C-Br bond

Anti-aromatic Aromatic

(c) Lewis acidity

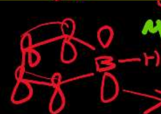


(d) Dipole moment



No polarization (0 dipole moment)

(c) ⇒



→ Empty

4π e⊖ Anti-aromatic so "B" can accept e⊖

6 πe⊖'s

Not Empty

involves in aromaticity

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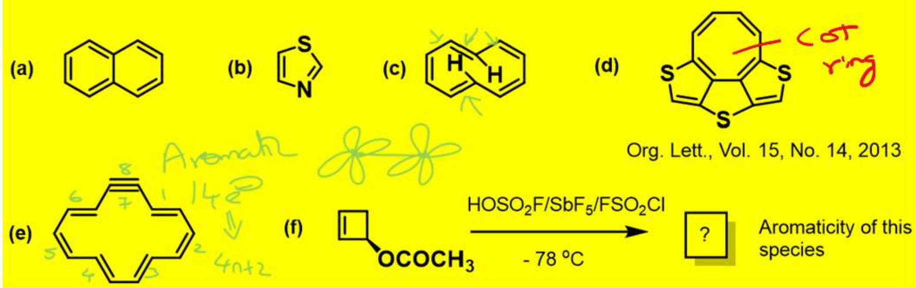
Stereochemistry

Problem Set 3

Q14. Classify the following molecules according to their aromaticity characteristics:



(a) 10 πe⊖ / cyclic / Planar ⇒ Aromatic

(b) 6 πe⊖ / Cyclic / Planar / Heteroatoms ⇒ Heteroaromatic

(c) Cyclooctatetraene part ⇒ Planarized due to ring fusion ⇒ COT ring has 8 πe⊖ / cyclic / Planar ⇒ ~~Anti~~ Non -aromatic

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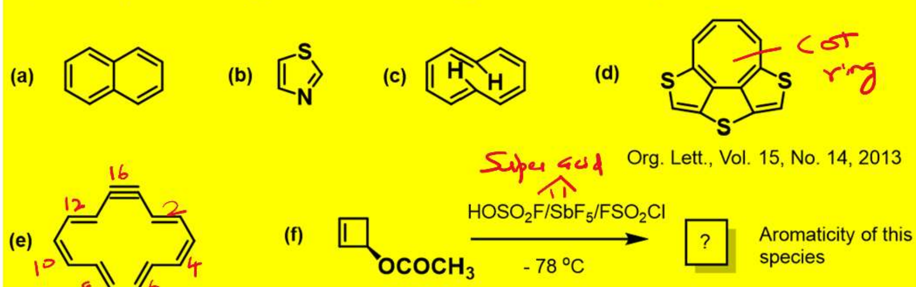
Stereochemistry

Problem Set 3

Q14. Classify the following molecules according to their aromaticity characteristics:



(e) 16 πe⊖ / 14 ⇒ part of ring πe⊖ / cyclic / Planar
4n+2 ⇒ Aromatic / 2 e⊖ ⇒ orthogonal alkyne

(f)

4n+2

✓ $\boxed{\text{Homoaromatic}}$ ⇐ Bypassed by sp³ center ⇐ Aromatic ✗

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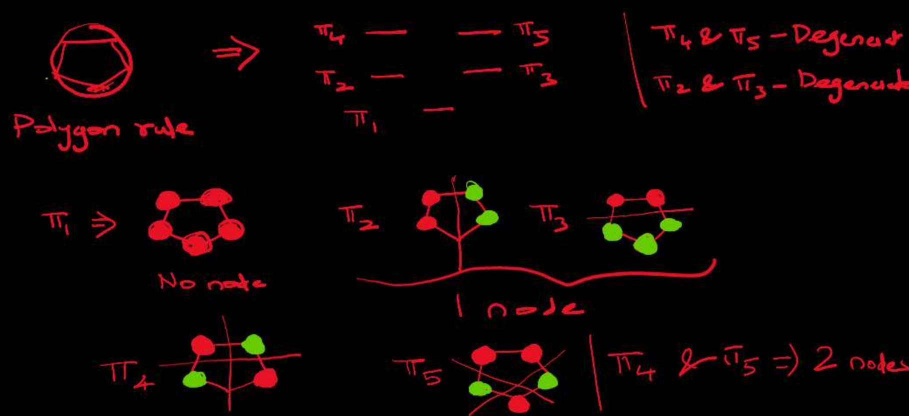
Stereochemistry

Problem Set 3

Q16. Draw the molecular orbitals (MO) diagrams of the following with appropriate energetic ordering:

(a) Cyclopentadienyl cation
(b) Cyclopentadienyl radical
(c) Cyclopentadienyl anion



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Stereochemistry

Problem Set 3

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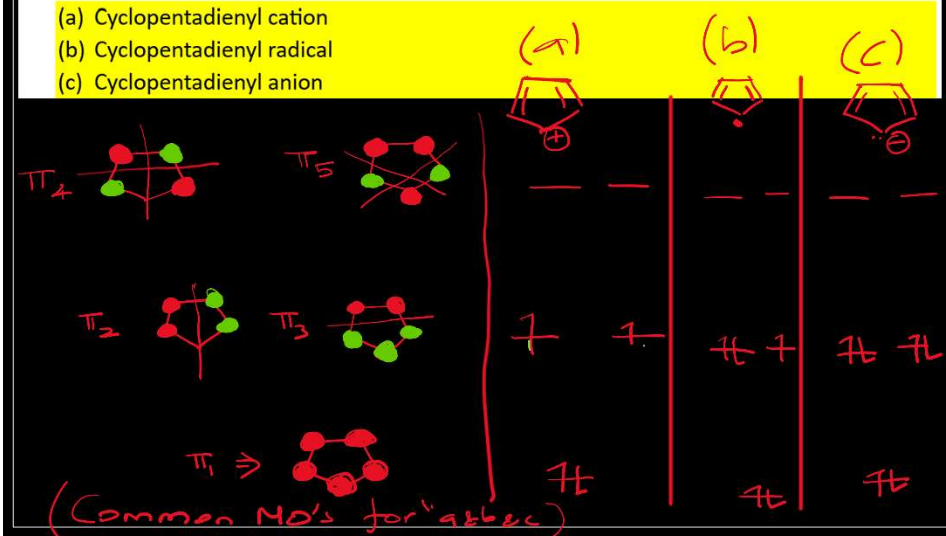
(a) Cyclopentadienyl cation
(b) Cyclopentadienyl radical
(c) Cyclopentadienyl anion



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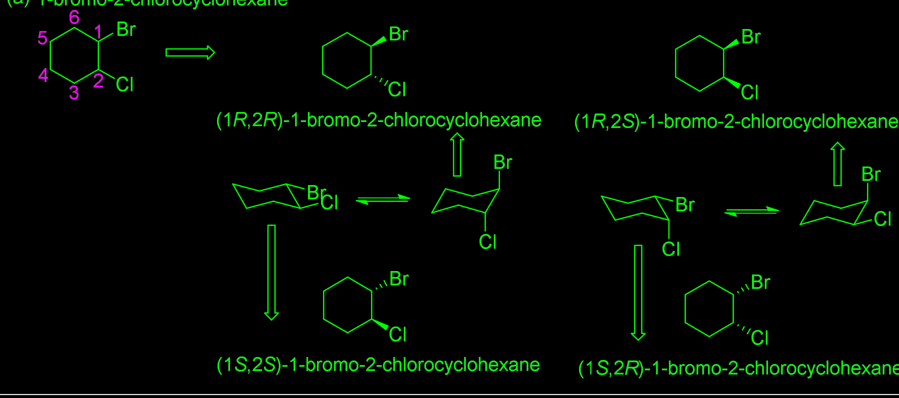
Stereochemistry

Problem Set 3

Q1. Draw all possible stereoisomers for each of the following compounds. State if no stereoisomers are possible.

a. 1-bromo-2-chlorocyclohexane
b. 1,2-dichlorocyclohexane
c. 1,2-dimethylcyclopropane

(a) 1-bromo-2-chlorocyclohexane



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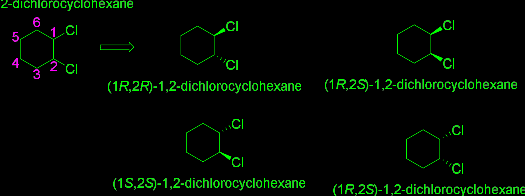
CHM102 Basic Organic Chemistry
Stereochemistry

Problem Set 3

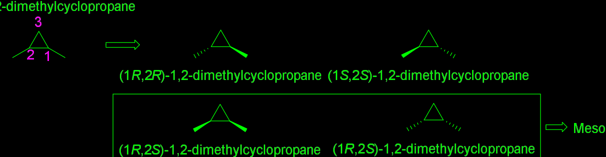
Q1. Draw all possible stereoisomers for each of the following compounds. State if no stereoisomers are possible.

a. 1-bromo-2-chlorocyclohexane
b. 1,2-dichlorocyclohexane
c. 1,2-dimethylcyclopropane

(b) 1,2-dichlorocyclohexane



(c) 1,2-dimethylcyclopropane



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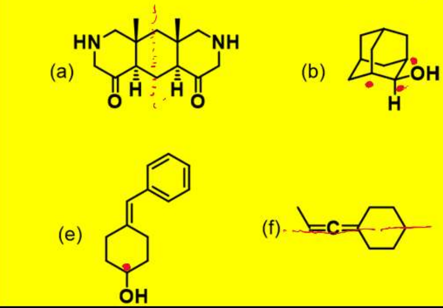
CHM102 Basic Organic Chemistry
Stereochemistry

Problem Set 3

Q2. Identify whether the following molecules are chiral. Also, specify whether the compound is enantiomer or diastereomer. If not, classify the compound as achiral, meso compound.



a, b, c & e ⇒ Point chirality
d – Axial chirality

(a) Plane of symmetry – Meso
(b) Chiral / Diastereomer
(c) Chiral / Enantiomer
(d) Axially chiral / Enantiomer
(e) Chiral / Enantiomer
(f) Plane of symmetry / achiral

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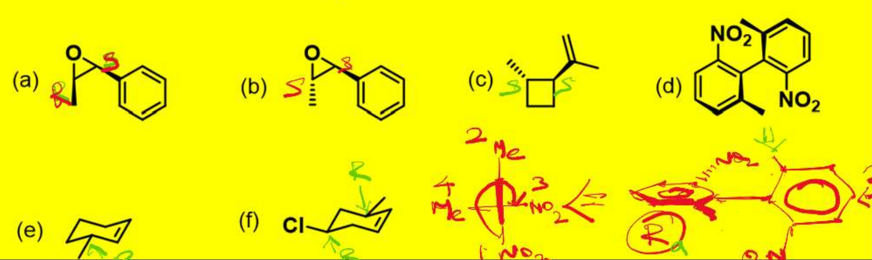
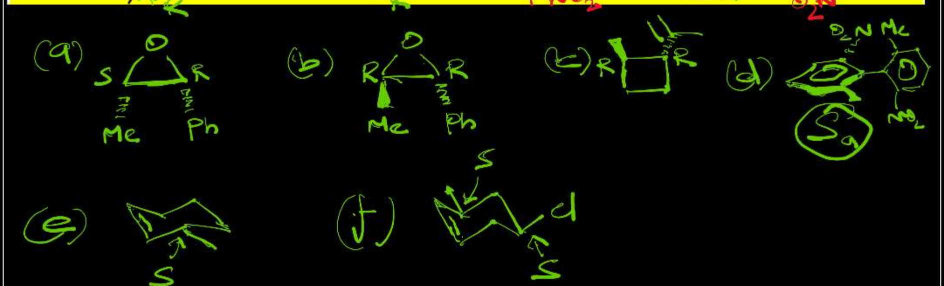
Stereochemistry

Problem Set 3

Q3. Draw enantiomers of the following stereoisomers.





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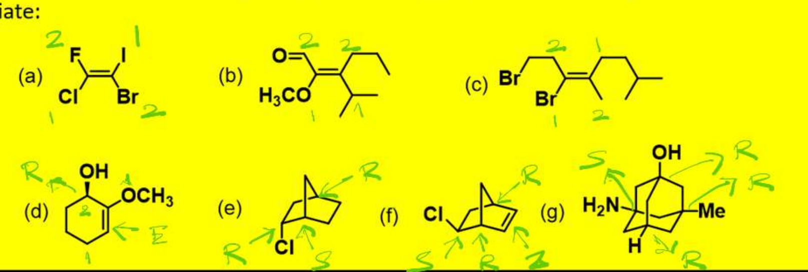
Stereochemistry

Problem Set 3

Q4. Name each of the following compounds using *R, S* and *E, Z* designations, wherever appropriate:



(a) E (b) Z (c) E

(d) R (OH group-C) & E (e) R, S, R

(f) S, R, Z, R (g) S, R, R, R

NH2 H Me OH ← Substitutions

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Stereochemistry

Problem Set 3

Q5. Which of the following compounds have an achiral stereoisomer?

(a) 2,3-dichlorobutane
(b) 2,3-dichloropentane
(c) 2,3-dichloro-2,3-dimethylbutane
(d) 1,3-dichlorocyclopentane
(e) 1,3-dibromocyclobutane
(f) 2,4-dibromopentane
(g) 2,3-dibromopentane
(h) 1,4-dimethylcyclohexane
(i) 1,2-dimethylcyclopentane
(j) 1,2-dimethylcyclobutane

Chiral ⇒ (a), (b), (d), (g), (j)

Achiral ⇒ (c), (e), (f), (h), (i)

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Stereochemistry

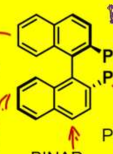
Problem Set 3

Q6. How many chiral centers are available in the following molecules? Indicate the chiral centers and/or type of chirality. Also, how many stereoisomers are possible for them?

Limonene Pinene Camphor Penicillin G

a – d ⇒ Point chirality
Bridge head → if chiral centers → cannot be inverted

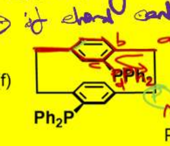
(e)



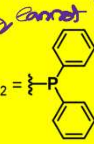
Ph = phenyl group

BINAP

(f)



PPh2 =

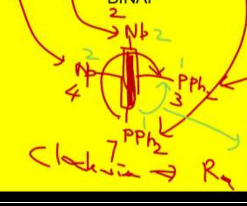


Ph = phenyl group

[2.2]-paracyclophane

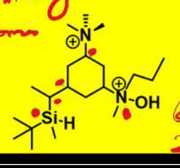
R_p

Planar chirality
2 isomers



Axial chirality
2 isomers

(g)



6 – chiral centers
$2^6 = 64$ isomers

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Stereochemistry

Problem Set 3

Q7. The enantiomers of 1,2-dimethylaziridine can be separated even though one of the "groups" attached to nitrogen is a lone pair. Explain.

Three membered-ring systems ⇒ Highly rigid
(inversion barrier for nitrogen lone-pair will be high)
⇓
So, the enantiomers can be separable

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Stereochemistry

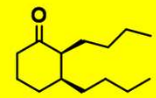
Problem Set 3

Q8. Whether the following molecule are chiral or optically active?

(a) 

(b) 

(c) 

(d) 

(a) ⇓ Center of Symmetry (or Inversion symmetry) ⇓ Achiral (optically inactive)

(b) ⇓ Plane of symmetry ⇓ Achiral (optically inactive)

(c) ⇓ Plane of Symmetry ⇓ Achiral (optically inactive)

(d) ⇓ Chiral (optically active)

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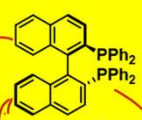
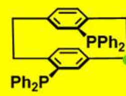
Stereochemistry

Problem Set 3

Q9. Identify *P* and *M* configurations (also R and S configurations) for the following compounds:

(a) 

Ph = phenyl group

(b) 

← P atom

Ph = phenyl group

Axis

Anticlockwise

Clockwise → R_a

M ← Helical chirality

← Pilot atom

Clockwise rotation

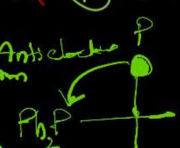
$a \rightarrow b \rightarrow c \rightarrow d$

⇓

R_p Isomer → CIP rules

Anticlockwise rotation

⇓



⇐

M Isomer / Helical chirality

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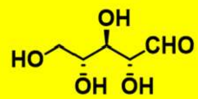
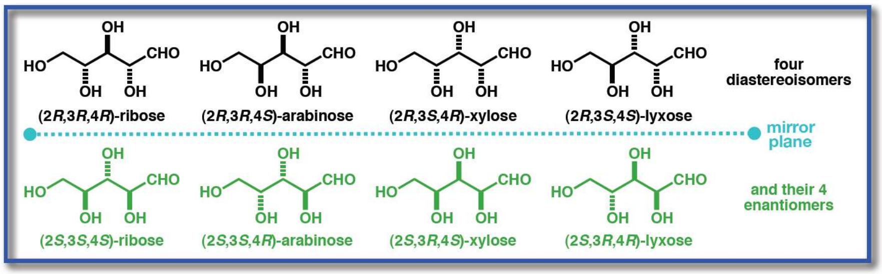
Stereochemistry

Problem Set 3

Q10. Draw the possible diastereomers and their enantiomeric pairs for the following molecule.



(2*R*,3*R*,4*R*)-2,3,4,5-tetrahydroxypentanal



- If a molecule has 3 stereogenic centres then it has potentially **8 stereoisomers** (4 diastereoisomers & 4 enantiomers)
- If a molecule has ***n*** stereogenic centres then it has potentially **2^n stereoisomers**

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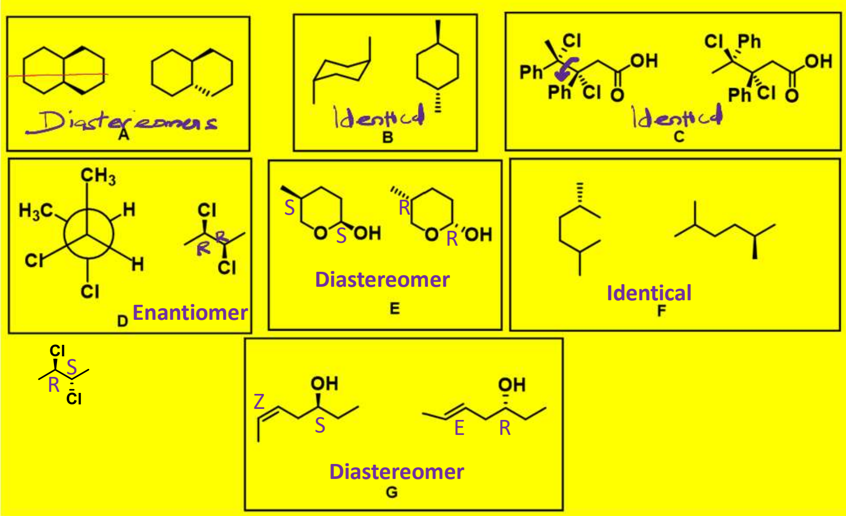
Stereochemistry

Problem Set 3

Q11. The relationship between the following pair of molecules among the possibilities "Identical/Constitutional isomers/Enantiomers/Diastereomers" is



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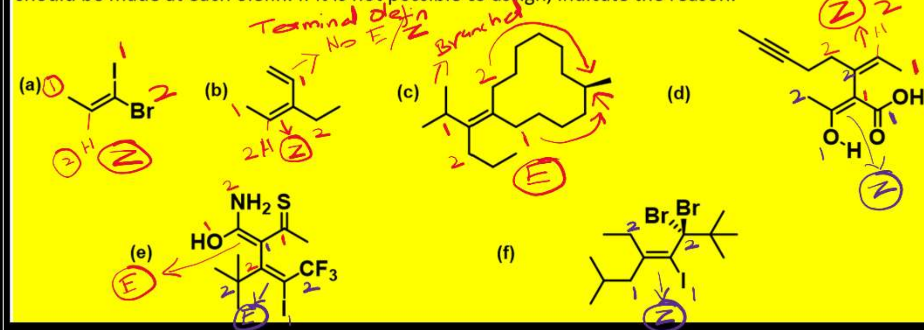
Stereochemistry

Problem Set 3

Q12. Assign absolute configurations (*E*/*Z*) to the following compounds. For poly-olefins, assignments should be made at each olefin. If it is not possible to assign, indicate the reason.



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